

Investigating Memory Hierarchy in Mobile and Gaming Devices: Apple M1 vs Playstation 5

Introduction

We live in an age of science and technology. In this modern era and with the rapid advancements of science and technology, modern computing devices are expected to provide high performance while also being efficient at the same time. For this reason, over the ages, we have seen the chip sizes getting shorter and shorter while also increasing performance at the same time within stricter power and thermal limits in order to ensure sustained performance over long periods of time. This is especially true for mobile systems and gaming consoles, where efficiency and performance are equally important in order to provide the user a good overall experience. A key factor behind this evolution has been the design of the memory hierarchy. In this report, we are going to explore how the Apple M1 and Playstation 5 structure their memory system in order to support demanding workloads such as gaming, multitasking, multithreading etc.

Background

Memory hierarchy is generally built on top of the idea of locality. Locality means programs usually tend to reuse recently accessed data stored in the cache or access data from nearby locations for better performance and short delays or latency (Patterson & Hennesy, 2017).

The Apple M1 is designed as a SOC with a unified memory architecture meaning that the CPU, GPU and other components in the architecture share the same memory pool, reducing the need to copy data between multiple components. The M1 chip consists of large L1 and L2 caches along with a system level cache in order to improve data reuse (Apple Inc., 2020). It uses LPDDR4X memory which is also optimized to ensure proper bandwidth flow while also being really battery efficient at the same time.

On the other hand, the Playstation 5, is built around a custom AMD architecture combining a Zen 2 CPU and RDNA 2 GPU (Freedman, 2020). The CPU consists of 8 cores and 16 threads where each CPU core includes separate L1 instruction and data cache and a private L2 cache, while the CPU cluster shares a larger L3 cache in order to ensure HPC (High Performance Computing). The PS5 also uses 16GB of GDDR6 memory connected via a wide 256-bit interface which ensures seamless delivery of very high bandwidth (Sony Interactive Entertainment, 2020).

Methodology

The analysis is based on official technical documentation, industry reports and academic references which are related to memory systems and is publicly available:

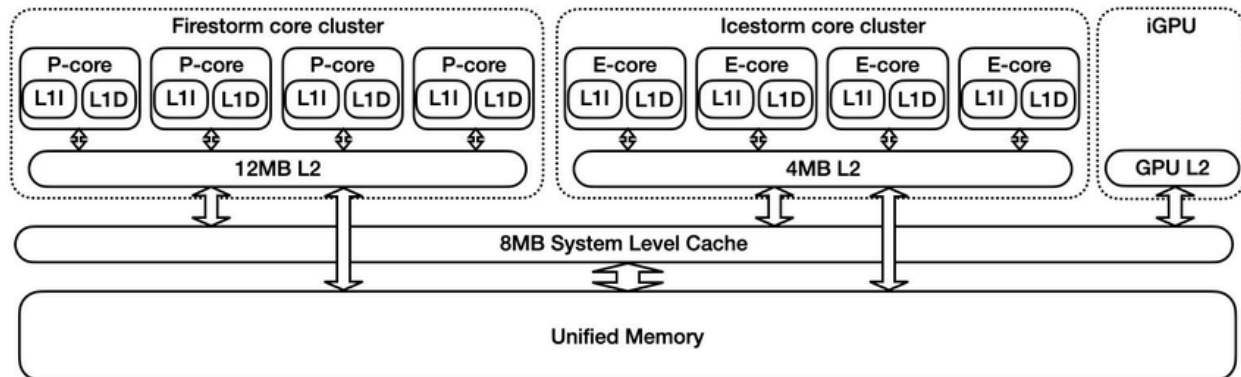


Figure: Cache Structure of Apple M1 (Xu et al., 2025)

Memory Hierarchy Comparison

Level	Apple M1	Playstation 5
Registers	CPU registers per core	CPU register per core
L1 Cache	Up to 192 KB / core	32 KB I + 32 KB D / core
L2 Cache	Up to 12 MB shared	512 KB/core
L3 / SLC	8 MB system level cache	16 MB shared
DRAM	LPDDR4X (approx 68 GB/s)	GDDR6 (approx 448GB/s)
Storage	NVMe SSD	NVMe SSD (custom)

Both Playstation 5 and Apple M1's system rely heavily on locality to improve overall performance. Data which is frequently used is kept in cache for easier and faster access, reducing the latency and also lifting the dependency to access slower memory locations.

The Apple M1 benefits from having a unified memory architecture which allows different processing units to access the same data without having to copy it over again and again. This

unified memory architecture reduces memory overhead and improves efficiency (Apple Inc, 2020).

On the other hand, the PS5 takes a different approach. It uses its high speed SSD to stream data dynamically which allows faster game loads and faster time for shaders to compile as well. This reduces pressure on the main memory.

Discussion

From the overall discussion, it is quite clear to us that the Apple M1 chip focuses on efficiency. Its unified memory architecture and large L1, L2 caches reduce the duplication of data between other components which ultimately results in lower power consumption. This makes it ideal for mobile devices where battery life is crucial to the user. The area that it sacrifices on is the memory bandwidth. The Apple M1 has a lower memory bandwidth compared to the PS5, which can limit performance in highly demanding graphics workloads.

The PS5, on the other hand, is optimized for raw non-negotiable performance. Its high bandwidth GDDR6 memory allows blazing fast data access while the custom SSD enables fast access to system files. This combo ensures smooth gameplay and high resolution graphics output but results in higher power consumption.

Therefore, a key tradeoff is noticed between the two systems. The M1 is energy efficient while being less powerful and on the other hand, the PS5 is high performance but also consumes more energy.

Lower power memory like the LPDDR4X used in the M1 uses less energy, making the M1 more sustainable over long term use. While the GDDR6 requires more power and cooling, resulting in improved performance but at a serious cost. So, all in all, designing a memory system that can balance out both performance and efficiency is essential for sustainable computing.

Conclusion

The memory hierarchy plays an important role in modern system designing. The Apple M1 shows how unified memory and proper cache usage can deliver strong performance while also maintaining lower power consumption. Again, the PS5 highlights how high bandwidth and performance can support the most demanding graphical workloads. As a result, it is clear that both designs show different design priorities tailored to the user's usecase. For someone, who is using a M1 macbook, battery life is crucial. On the other hand, for someone who is using the PS5, smooth gameplay is crucial. So, to conclude, future design may combine these approaches in order to create a balanced experience for both high performance and efficiency.

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